

Flow-Factory OPD & Ensemble Eval Experiments

Flow-Factory

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1 Experimental Setup

Offline `ensemble-eval` on SD3.5 Medium: FlowGRPO LoRA checkpoints are fused each denoising step, then evaluated on `geneval`, `pickscore`, and `ocr` held-out sets. Checkpoints: `jieliu/SD3.5M-FlowGRPO-Text (OCR)`, `jieliu/SD3.5M-FlowGRPO-PickScore`, `jieliu/SD3.5M-FlowGRPO-GenEval`. Multi-teacher rows use AVG unless marked PCGRAD. Table 1: **bold** = single-teacher in-domain; gray = multi-teacher in-domain; red/blue = change vs. max single-teacher baseline in that row. OPD rows (*peak*) use the maximum eval logged so far. Route: ✓ = `teacher_route_by_source`; × = off or not teacher-specific (e.g. `pw-avg`).

2 Main Results

Table 1: Cross-benchmark `eval/` means. Top: offline ensemble-eval (last eval). Bottom: OPD training runs still **running** (peak eval). **Bold**: single-teacher in-domain.

Config.	Route	Agg.	GenEval benchmark		PickScore benchmark	OCR benchmark	
			GenEval	PickScore	PickScore	OCR	PickScore
<i>Ensemble eval (offline, last eval)</i>							
SD3.5 (base)	—	—	0.657	0.875	0.835	0.574	0.866
geneval	—	—	0.937	0.888	0.829	0.656	0.862
ocr	—	—	0.685	0.876	0.837	0.931	0.869
pickscore	—	—	0.641	0.918	0.895	0.690	0.915
geneval+ocr	—	AVG	0.850-0.087	0.888-0.000	0.837+0.000	0.875-0.055	0.874+0.005
geneval+pickscore	—	AVG	0.830-0.107	0.908-0.010	0.873-0.021	0.675-0.015	0.899-0.017
ocr+pickscore	—	AVG	0.680-0.005	0.901-0.017	0.873-0.022	0.865-0.066	0.899-0.016
geneval+ocr+pickscore	—	PCGRAD	0.794-0.143	0.901-0.017	0.864-0.031	0.844-0.087	0.894-0.022
<i>OPD training (running, peak eval)</i>							
pw-avg	×	AVG peak	0.787-0.150	0.898-0.019	0.861-0.033	0.830-0.101	0.893-0.023
pw-sum	×	SUM peak	0.755-0.182	0.897-0.021	0.859-0.035	0.808-0.122	0.892-0.023
pw-sum	✓	SUM peak	0.855-0.082	0.891-0.026	0.878-0.016	0.898-0.033	0.883-0.032
v-pcgrad	×	PCGRAD peak	0.756-0.181	0.897-0.021	0.858-0.036	0.805-0.126	0.892-0.024
v-pcgrad	✓	PCGRAD peak	0.834-0.103	0.890-0.028	0.876-0.018	0.895-0.035	0.884-0.032
diff-opd	✓	D-OPD peak	0.913-0.024	0.892-0.026	0.887-0.008	0.917-0.013	0.884-0.032

3 Full Eval Metrics

Table 2: All scalar `eval/` metrics (std and GenEval per-tag). Ensemble: last eval; OPD: peak.

Config.	Agg.	GenEval benchmark														PickScore benchmark		OCR benchmark	
		GenEval							PickScore							PickScore	OCR	PickScore	
		Std	Color attr	Colors	Counting	Position	Single obj	Two obj	Std	Color attr	Colors	Counting	Position	Single obj	Two obj	Std	Std	Std	
Ensemble eval (offline, last eval)																			
SD3.5 (base)	—	0.395	0.521	0.711	0.761	0.450	0.844	0.636	0.042	0.900	0.879	0.875	0.866	0.855	0.873	0.057	0.393	0.039	
geneval	—	0.205	0.854	0.867	0.994	0.950	1.000	0.982	0.038	0.920	0.890	0.890	0.869	0.864	0.887	0.056	0.346	0.038	
ocr	—	0.575	0.531	0.689	0.802	0.490	0.867	0.664	0.042	0.902	0.882	0.879	0.874	0.856	0.874	0.057	0.173	0.041	
pickscore	—	0.407	0.479	0.622	0.913	0.440	0.800	0.655	0.043	0.949	0.919	0.917	0.926	0.881	0.921	0.058	0.350	0.039	
geneval+ocr	Avg	0.307	0.719	0.800	0.898	0.730	0.911	0.945	0.040	0.921	0.886	0.889	0.880	0.867	0.884	0.056	0.231	0.040	
geneval+pickscore	Avg	0.319	0.740	0.800	0.904	0.770	0.933	0.873	0.041	0.947	0.907	0.907	0.906	0.875	0.909	0.057	0.349	0.038	
ocr+pickscore	Avg	0.387	0.490	0.711	0.867	0.550	0.778	0.682	0.042	0.935	0.897	0.903	0.905	0.869	0.900	0.057	0.253	0.040	
geneval+ocr+pickscore	PCGRAD	0.338	0.625	0.844	0.959	0.650	0.867	0.818	0.041	0.940	0.899	0.898	0.898	0.870	0.900	0.058	0.259	0.039	
OPD training (running, peak eval)																			
pw-avg	Avg peak	0.393	0.677	0.800	0.957	0.760	0.889	0.873	0.042	0.938	0.898	0.900	0.900	0.869	0.898	0.058	0.393	0.040	
pw-sum	Sum peak	0.395	0.604	0.800	0.935	0.710	0.867	0.836	0.042	0.935	0.896	0.898	0.896	0.869	0.897	0.058	0.388	0.040	
pw-sum	Sum peak	0.395	0.812	0.889	0.930	0.780	0.933	0.873	0.042	0.923	0.900	0.893	0.883	0.867	0.886	0.059	0.388	0.042	
v-pcgrad	PCGRAD peak	0.395	0.573	0.800	0.930	0.710	0.844	0.836	0.042	0.934	0.895	0.897	0.897	0.869	0.897	0.058	0.388	0.040	
v-pcgrad	PCGRAD peak	0.395	0.781	0.867	0.913	0.740	0.933	0.855	0.042	0.925	0.899	0.892	0.882	0.864	0.883	0.059	0.388	0.042	
diff-opd	D-OPD peak	0.389	0.833	0.889	0.970	0.890	0.956	0.955	0.042	0.925	0.900	0.895	0.885	0.870	0.890	0.059	0.388	0.041	